

Axons of Individual Dorsal Horn Neurons Bifurcated to Project in Both the Anterolateral and the Postsynaptic Dorsal Column Systems

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Abstract—Sensory information stimulates receptors of somatosensory system neurons generating a signal that codifies the characteristics of peripheral stimulation. This information reaches the spinal cord and is relayed to supra-spinal structures through two main systems: the postsynaptic dorsal column-medial lemniscal (DC-ML) and the anterolateral (AL) systems. From the classical point of view, the DC-ML has an ipsilateral ascending pathway to the Gracilis (GRA) or Cuneate (CUN) nuclei and the AL has a contralateral ascending pathway to the ventral posterolateral (VPL) thalamic nucleus. These two systems have been the subject of multiple studies that established their independence and interactions. To analyze the ascending projections of L1–L5 spinal dorsal horn neurons in the rat, two retrograde neuronal tracers were injected into the GRA and the VPL. Additionally, an electrophysiological study was performed by applying electrical stimulation at the GRA or VPL and recording antidromic evoked activity in single unit spinal cord cells. Importantly, a subset of spinal dorsal horn neurons exhibited double staining, indicating that these neurons projected to both the GRA and the VPL. These double-stained neurons were located on both sides of the dorsal horn of the spinal cord. The spinal dorsal horn neurons exhibited antidromic and collision activities in response to both GRA and VPL electrical activation. These results show spinal cord neurons with bifurcated bilateral projections to both the DC-ML and AL systems. Based on these results, we named these neurons bilateral and bifurcated cells. © 2017 IBRO. Published by Elsevier Ltd. All rights reserved.

Key words: retrograde tracers, collision test, bilateral and bifurcated cells, spinal projecting cells, anatomical and electrophysiological study.

INTRODUCTION

Skin receptors are specialized to detect particularities in skin stimulation and transmit this coded information to spinal dorsal horn neurons. These neurons, also called second order neurons, send the information to the supra-spinal structures by two main pathways: the

ipsilateral dorsal column-medial lemniscal (DC-ML) and the contralateral anterolateral (AL) systems.

The DC-ML system senses primary afferents concerning tactile and proprioceptive information arriving at the spinal cord and forms the ipsilateral dorsal column pathway to reach the Gracile (GRA) or the Cuneate (CUN) nuclei. Then, neurons at the GRA and CUN reach the contralateral ventral-postero-lateral (VPL) and ventral-postero-medial thalamic nucleus via the medial lemniscus pathway. Also, primary afferents reach another cluster of spinal neurons called postsynaptic dorsal column, that send their axons to the ipsilateral GRA or CUN nucleus and then by the medial lemniscus reach the contralateral thalamus (Angaut-Petit, 1975a,b).

The AL system receives primary afferent information concerning pain, temperature and crude touch, and dorsal spinal cord projection neurons extend their axons contralaterally to reach the brain stem reticular formation, midbrain, tectal region and several thalamic

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Abbreviations: AL, anterolateral system; anti-anti, double antidromic activated cell; Anti, antidromic activated cell; AV, anterior ventral thalamic nucleus; CUN, Cuneate nuclei; DC-ML, postsynaptic dorsal column-medial lemniscal system; EEG, electroencephalographic activity; FG, Fluoro Gold retrograde tracer; FR, Fluoro Rubi retrograde tracer; GRA, Gracilis nuclei; LDV, lateral dorsal ventral thalamic nucleus; Po, posterior thalamic nucleus; RT, reticular thalamic nucleus; STT, spinothalamic tract; TB, True Blue retrograde tracer; VL, ventral lateral thalamic nucleus; VPL, ventral posterolateral thalamic nucleus; WDR, wide dynamic range neurons; ZI, zona incerta.

structures (Martin and Jessell, 1991; Guilbaud et al., 1993; Gardner and Johnson, 2013). Hence, the AI system is formed by the spinoreticular, spinomesencephalic and the spinothalamic (STT) tracts. Two reviews on pain and the somatic and visceral pathways by Almeida et al. (2004) and Millan (1999) summarize the classical view of these tracts.

The DC-ML and the STT systems have been described as independent and separate pathways until they reach the thalamus, and little is known about their interconnections, which are the main focus of the present study. In pioneer studies, Kevetter and Willis (1984) described the 2% of spinothalamic cells projecting to both sides of the thalamus. Also, these authors described that the 38% of spinothalamic cells project to the ipsilateral medial thalamus and the 19% of cells project to the ipsilateral lateral thalamus. Lu and Willis (1999) reviewed this work with the same precision. Also, Apkarian and Hodge (1989) and Hodge and Apkarian (1990) described that 17% of the spinal projecting cells reach the ipsilateral thalamus.

As well as the DC-ML and AL systems (Hendry et al., 1999), yet we found no reports concerning their interactions or interconnections. Therefore, the present study describes a particular set of spinal cord dorsal horn neurons projecting to both ipsilateral GRA and contralateral VPL. To the best of our knowledge, this is the first description of interconnections between the DC-ML and AL systems.

Retrograde fluorescent tracers were injected into the left GRA and the right VPL thalamic nuclei. The results show that both fluorescent tracers labeled individual neurons on both sides of the spinal cord (L1–L5). To confirm our results, we performed the electrophysiological collision test using electrical stimulation at the GRA and/or VPL to produce antidromic responses in neurons of the spinal cord identified as wide dynamic range (WDR).

Preliminary results were presented in an IASP congress (Condés-Lara et al., 2016).

EXPERIMENTAL PROCEDURES

The experiments were carried out in accordance with the National Institute of Health Guide for the Care and Use of Laboratory Animals (NIH Publications No. 80-23) revised 1996, as well as the European Communities Council Directive of 24 November 1986 (86/609/EEC). Also, the recommendations of the IASP Ethical Guidelines (Zimmermann, 1983) and the recommendations of the Instituto de Neurobiología Ethical Commission. The Animal House of our institute provided male Sprague–Dawley rats weighing an average of 280–310 g. They were individually housed in a temperature-controlled (23 ± 3 °C) environment and subjected to a 12 h/12 h light/dark cycle with food and water available *ad libitum*. Two sets of experimental approaches were used: Retrograde fluorescent tracers, with three tracers' combinations: True Blue (TB) in the GRA and Fluoro Gold (FG) in the VPL; TB in the GRA and Fluoro Rubi (FR) in the VPL; and TB in the GRA and FG in the VPL with immunofluores-

cence. The second experimental approach used Electrophysiology techniques.

Note that while describing results, observations of the left side were considered ipsilateral (left GRA) and observations on the right are considered contralateral (right VPL). The reason was that our electrophysiological recording sites in the spinal cord of the rat were the left side (ipsilateral) and the receptive field evoking responses were also the left side (ipsilateral).

Injection of fluorescence retrograde tracers in the GRA and VPL

Injection of Fluoro Gold (FG) and True blue (TB). Fifteen rats were anesthetized with a ketamine/xylazine mixture (70/6 mg/kg, i.p.) and placed in a stereotaxic frame. The animals head was initially secured in ventroflex position. A skin incision was made and then the muscles of the back of the neck were dissected laterally from the midline. The atlanto-occipital membrane and underlying meninges were opened, exposing the cisterna magna and the dorsum of the lower medulla. A stainless-steel microelectrode carrying a small pellet (400–500 μ m) of True Blue (TB) (True Blue Chloride, No. Cat T-0695 Sigma, St. Louis, MO USA) attached to the tip was inserted into the left GRA at 0.5 mm lateral to the middle of the caudal region of the nucleus (between 0.5 and 1.0 mm caudal to the obex) under a microscope. Then, with the head in a stereotaxic position (Paxinos and Watson, 1998), a trephine hole was drilled into the skull to pressure inject 60–90 nl of a Fluoro Gold (FG) (Fluoro-Gold, Fluorochrome, LLC, Denver, CO, USA) solution (2.5% in distilled water) into the right VPL. The glass micropipettes were prepared under a microscope such that they had a 4–10 μ m tip; the pipettes were attached to a 1 μ l Hamilton syringe mounted in a microdriver system. The pipettes and syringe were filled with glycerin and only the pipette tip was charged with the tracer. This procedure allowed us to perform injections of $20\text{--}30\text{ nl} \pm 10\text{ nl}$.

The FG injections were performed at three depths for each experimental animal with the purpose of making an extended dorsal ventral VPL injection (interaural stereotaxic coordinates: AP 6.1; L 2.8; H 3.3, 3.5 and 3.7 mm) (Paxinos and Watson, 1998).

The animals were allowed to recover for 13 days. Then, an overdose of anesthesia (50 mg/kg of pentobarbital) was delivered, and the animals were perfused through the heart using a peristaltic pump with a brief rinse of saline (200 ml), followed by 4% paraformaldehyde (300 ml) in 0.1 M phosphate buffer (pH 7.4). The brains and spinal cords were removed, postfixed for an additional 2 h, and cryoprotected in a 30% sucrose solution 0.1 M phosphate buffer (pH 7.4) for two days. The brains and spinal cords were cut (40 μ m) in a freezing microtome. Spinal cord sections were collected in 5 serialized series of 50 to 65 sections each one. The first series was used to observe the spinal cord location, the second and third were used for fluorescence process and the other two stored for

further verifications. Each section had 40 μm and the distance between section was 200 μm giving us about 64 spinal cord sections per series in each rat. All sections were then mounted on gelatinized slides, dehydrated, cleaned, and cover slipped for the subsequent microscopic analysis. The brain and spinal cord sections were observed under an Olympus IX81 microscope equipped with an OH-ORCA-FLASH40 digital camera and fluorescence illumination system or with a confocal microscope (Zeiss Axiovert LSM 500 meta) with a multiphotonic laser (Chameleon ultra) and the proper specific filter for each fluorescence wavelength spectrum, and digital images of the labeled neurons were obtained.

Labeled cells count. One of the above described serial sections was used to count the number of stained cells. With a total of 320 spinal cord sections between L1 and L6 taken from 6 rats (64 per animal) we counted the stained dorsal spinal cord cells located in both the ipsilateral and contralateral sides. These counts were used to construct Table 1. The profiles of selected sections and the location of labeled neurons were analyzed and quantified using the Data Image Tracer (version 1.2) software, as well as a semi-automatic Image Processing System MD3 (Minnesota Datametrics). Computer-assisted images from adjacent Nissl and immunofluorescence sections were generated using a Leica DMLB. Digital higher-power images (100 $\times$ and 200 $\times$) of labeled neurons were also obtained with this same device. Plots of labeled neurons were placed on the immunofluorescence sections and on spinal cord diagrams from Paxinos and Watson (1998) atlas.

Fluoro Ruby (FR) and True blue (TB). Five rats were prepared to compare the results obtained using the FG and TB injections. The purpose of this maneuver was to confirm that neurons were double marked. The tissue sections, sites of injections and spinal cord samples were similar to those described previously. Briefly, small pellets of TB were inserted into the left GRA nucleus under a microscope, and 60 to 90 nl of a 10% solution of FR (Fluoro-Ruby No. Cat. AG335, Millipore, Temecula, California, USA) in saline was injected at three different depths in the right VPL at the same coordinates described above. The animals were allowed to recover for 10–13 days; then they were deeply anesthetized and perfused as described in the previous section. The tissue sectioning, mounting and analysis were similar to the previous section.

Immunofluorescence for FG. To ensure that the FG-stained cells did indeed co-localize with TB, other sections of the same rats were preincubated with blocking serum for 20 min at room temperature and then incubated overnight at 4 °C with rabbit anti-FG (Fluorochrome, LLC) diluted 1:500 in 0.1 M phosphate-buffered saline (PBS) (pH 7.4) containing 0.5% Triton X-100 (Triton X-100, No. de cat. X100, Sigma–Aldrich, St. Louis, MO USA) and 5% normal goat serum (Goat serum, No. de cat. 16,210, Gibco, Invitrogen, Auckland, New Zealand). After rinsing in PBS, the sections were incubated with a secondary antibody, AlexaFluor 555-conjugated goat anti-rabbit diluted 1:300 (AlexaFluor 555-conjugated goat anti-rabbit, No. de cat. A21428, Molecular Probes by Life technologies, Eugene, OR, USA), for 24 h. After rinsing, the sections were mounted on gelatin-coated, dehydrated, cleared, and cover slipped with DPX. The sections were examined under a confocal microscope (Zeiss Axiovert LSM 500 meta) with a multiphotonic laser (Chameleon ultra). Specific filters were used for each fluorescence wavelength, and digital images of the labeled neurons were obtained.

Electrophysiological recordings of the spinal cord cells activated by electrical stimulation of the GRA and VPL

Twenty-five male Wistar rats (280–310 g) were housed individually under 12-h:12-h light/dark conditions, with food and water available *ad libitum*. The rats were anesthetized with 3% sevoflurane in an atmosphere of 1/3 oxygen and 2/3 nitric oxide. The rats were placed in a hermetically sealed box with the anesthesia and when a loss of reflexes was confirmed, they were tracheotomized to provide artificial ventilation and placed in a stereotaxic frame to fix their heads and spinal cords.

Concentric bipolar electrodes were placed in the right VPL at AP 6.1 ± 0.5 ; L 2.8 ; and H 3.5 ± 0.3 mm using the interaural point according to the Paxinos stereotaxic atlas (Paxinos and Watson, 1998), and the atlanto-occipital foramen was exposed to allow access to the left GRA nucleus. The spinal cords were exposed at the L3–L5 level to record the neuronal activity on the left side.

The dorsal spinal cord cells were recorded using glass micropipettes filled with 4% pontamine sky blue in 1 M KCl (impedance of 8–10 M ohms). At the end of the recordings or in the recording sites of interest, pontamine blue was injected iontophoretically using a cathodic current (15–20 μA) for 25 min. The cells were

Table 1. Single and double stained dorsal horn cells number. The Table shows the number of spinal cord cells stained with only TB or FG and with both TB and FG. The data were obtained from 6 rats. The mean of ipsilateral and contralateral cells is also shown. Note the differences between the mean number of TB/FG labeled ipsilateral (2.6) and contralateral cells (0.58). Ipsilateral refers to the left side and contralateral refers to the right side because we recorded the spinal cord cells on the left side

	Ipsilateral		Contralateral	
	Number	Mean	Number	Mean
TB (only)	2302	1.429	1264	0.776
FG (only)	1352	0.839	913	0.556
TB/FG	4365	2.684	954	0.585

recorded using a Dual Microprobe Amplifier WPI without filters at an amplification of X10 and then the signals were amplified (X100) in a Grass P511 (the bandpass filters were set between 300 and 3000 Hz). Another Grass P511 was used to amplify the electroencephalographic (EEG) activity from two electrodes placed at the cortex. The EEG was used to indicate the level of anesthesia and was characterized by the presence of slow waves sleep (2–6 Hz); in this case, the bandpass filters were fixed at 3–300 Hz. The activities were digitalized using a Spike 2 and the appropriate software (Cambridge Electronic Device, CED).

We recorded 225 cells at the L3–L4 segments of the spinal dorsal horn that have a receptive field on the left hind foot. Only the wide dynamic range cells that responded to tactile and nociceptive stimulation were tested. Under these conditions, 172 cells were tested by electrically stimulating the GRA or VPL. The response latencies were used to calculate the conduction velocities. In addition, for the recorded spinal cord cells we considered a distance of 85 mm from the GRA and 100 mm from the VPL.

The recorded cells were found during the microelectrode displacement into the spinal cord and were activated by gently taping into the receptive field (RF). When a neuron was found in order to characterized the cell responses, mechanical and nociceptive stimulus (taping the receptive field with different intensities and hand squeezing) was delivered. Finally, a pair of electrodes (fine needles 27 G) attached to a stimulus isolator unit were inserted into the RF to deliver electric stimulation into the receptive field of the recorded neuron. We constructed peri-stimulus time histograms after triggering the activity by analyzing the RF signal artifact. We also used the spontaneous or evoked action potentials to trigger the GRA or VPL electrical stimulation with variable delays. These delays in the GRA or VPL stimulation allowed us to measure the collision time between the spontaneous and evoked action potentials in response to GRA or VPL stimulation in the recorded spinal cord cells and to describe their antidromic characteristics. The antidromic spike responses, evoked by stimulating the contralateral VPL or ipsilateral GRA, were provoked by single 1-ms square-wave pulses with intensities starting at 100 μ A and increasing until a threshold was reached for each cell, but never exceeding 300 μ A. When a stable cell recording was obtained, the following criteria were used to establish the antidromic characteristic of the cell responses: a constant latency, the ability to follow a stimulus train of 333 Hz, and a collision of the orthodromic spikes with the antidromically evoked spikes. The collision test was performed by first testing the GRA or VPL stimulation. If we observed a spike response, we used the spontaneous activity or spikes evoked by the RF stimulation to trigger the GRA or VPL stimulation with variable delays to observe the collision between the triggering spike and the antidromic GRA- or VPL-evoked response. The collision time was calculated as the sum of the latency of the antidromic response

plus the refractory period. The latencies and conduction speeds were expressed as the means $\pm$ the standard error of the mean (SEM). At the end of the experiments, the recording sites were marked by electrophoretic deposition of pontamine sky blue (20 min 15 μ A cathodic current) and electrolytic lesions were made at the stimulating points, then an overdose of anesthesia was delivered. The lumbar spinal cords and the brains were removed and fixed by immersion in a 10% formaldehyde solution for 72 h. Samples were frozen, cut in serial 40- μ m sections, and Nissl-stained with cresyl violet. GRA and VPL electrolytic lesions were confirmed by localizing the stimulating electrode tips.

RESULTS

We will first describe our histological results and then our electrophysiological results. We only analyzed the animals in which the FG injection into the VPL thalamic nucleus and the TB injection into the GRA nucleus were located in the appropriate sites, as well as the cases in which we switched FG for FR to corroborate our results. We also used the FG antibody and an AlexaFluor 555-conjugated secondary antibody to verify our results. In Fig. 1, we can observe the histological location of the TB dye in the GRA, as well as the retrograde staining of the cells in the CUN after the FG injection into the VPL.

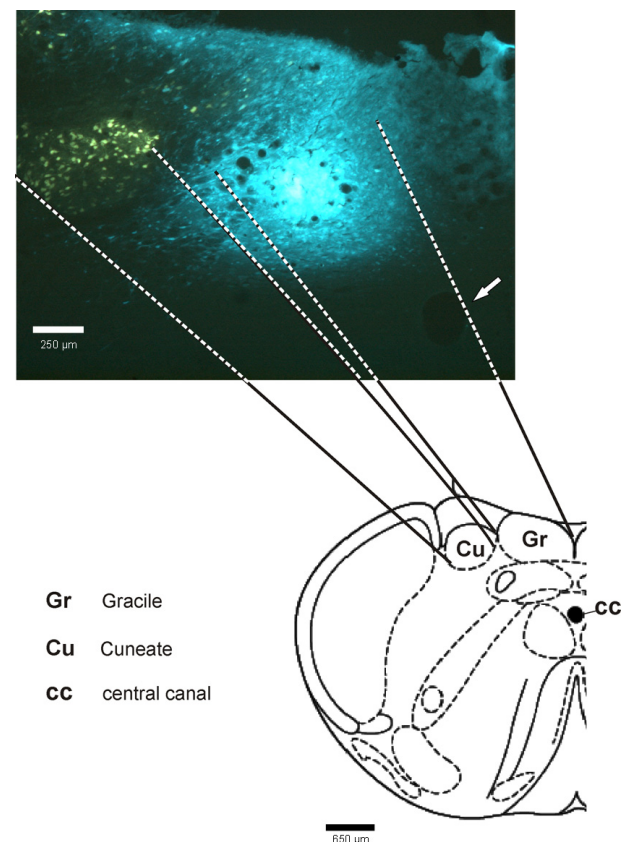


Fig. 1. Histological section showing the TB deposit site in the GRA. Observe the retrograde labeling of cells in the CUN nucleus after the FG injection in the VPL. The lower schema locates the structures corresponding to the section.

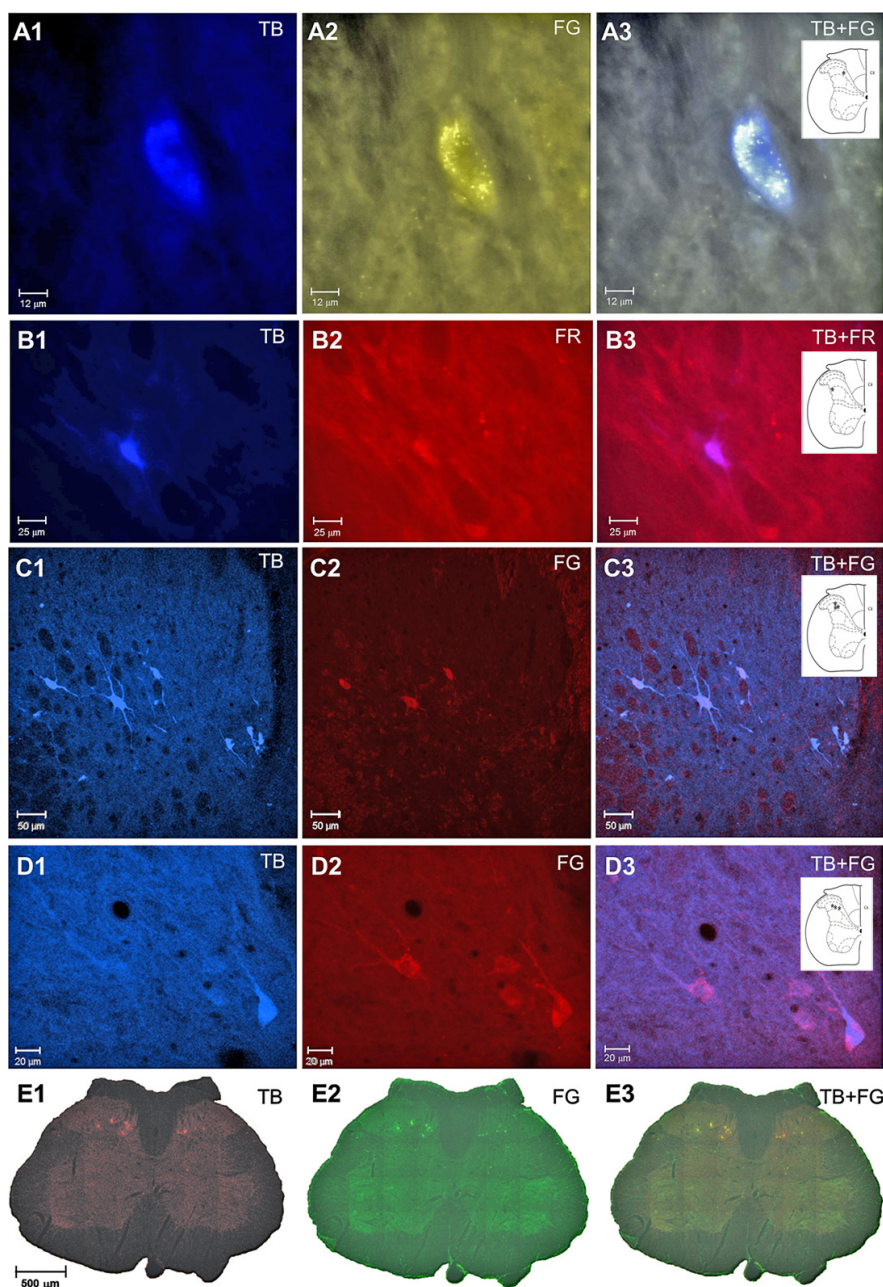


Fig. 2. Labeled spinal dorsal horn cell. The first column shows images of the labeled cells after the TB injection in the GRA, the middle column shows images of the cells projecting to the VPL, and the right column shows the merged images of the cells projecting to both the GRA and VPL. TB was inserted into the GRA nucleus, and FG or FR B2 was injected into the VPL. The first line A1, 2 and 3 shows a cell labeled with TB, FG and TB/FG located in the medial portion of the Rexed layer 4. The second line shows an example of a cell labeled with TB showed in B1, FR in B2 and both TB and FR in B3. The TB- and FR-positive cells were located in the lateral portion of layer 5; note the change in the location of the FR tracer. Lines C1, 2 and 3 illustrate the confocal microscopy images of spinal cord cells at layer 4 using the FG antibody. The image in C1 illustrates the TB-labeled cells, and the image in C2 shows the FG antibody-positive cells. The single and double TB and FG antibody-labeled cells are displayed in lines D1, 2 and 3. The images were obtained from the medial portion of spinal cord layer 4 using a confocal microscope. The E1, 2 and 3 images were obtained from a spinal cord mosaic using a confocal microscope. The images show the TB-labeled E1, FG-labeled cells E2 and FG/TB double-labeled cells E3 in layer 4 of the dorsal horn on both sides of the spinal cord. Note that the color of the TB-labeled cells was changed to red for better contrast. The spinal cord scheme insertions in A3, B3, C3 and D3 indicate the cell's locations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Due to the high levels of TB staining, we could not observe the FG retrograde staining of the cells in the GRA. Nevertheless, the image in Fig. 1 confirms that TB was present in the GRA and that the FG-stained cells in the CUN indicate retrograde FG transport from the VPL.

We selected TB and the FG or FR retrograde tracers' due to their different excitation wavelengths, which allows selective and unambiguous identification.

TB- and FG-positive neurons in the spinal cord

The TB and FG applications stained the spinal dorsal horn cells between L1 and L6 with a variable distribution. At L1 and L6, the stained cells were very scarce; in contrast, at L3 and L4 we found a peak in the cell density. In fact, we observed cells that were stained with TB, FG and TB/FG, as illustrated in Fig. 2A1, A2 and A3. The cells were bilaterally distributed, but there were fewer cells on the right side of the spinal cord, even though FG was injected into the right VPL and TB into the left GRA. From sections between L1 and L6 taken from 6 rats, the mean number of stained cells per section was 1.42 ipsilateral TB- (only) positive cells vs 0.77 contralateral TB-(only) positive cells, and 0.83 ipsilateral FG-(only) positive cells vs 0.55 contralateral FG-(only) positive cells. We observed 2.68 double-stained cells (TB/FG) on the ipsilateral side and 0.58 on the contralateral side (a mean rate of 4.62 cell ipsilateral by 1 cell contralateral). Subsequently, we plotted the number of cells on each side (Table 1) and observed differences between sides using the mean number of cells/side.

TB- and FR (Fluoro Ruby)-labeled neurons in the spinal cord

Injecting FR instead of FG into the VPL confirmed the previous results. Fig. 2B1, B2 and B3 shows the single- and double-labeled cells that were positive for TB and/or FR.

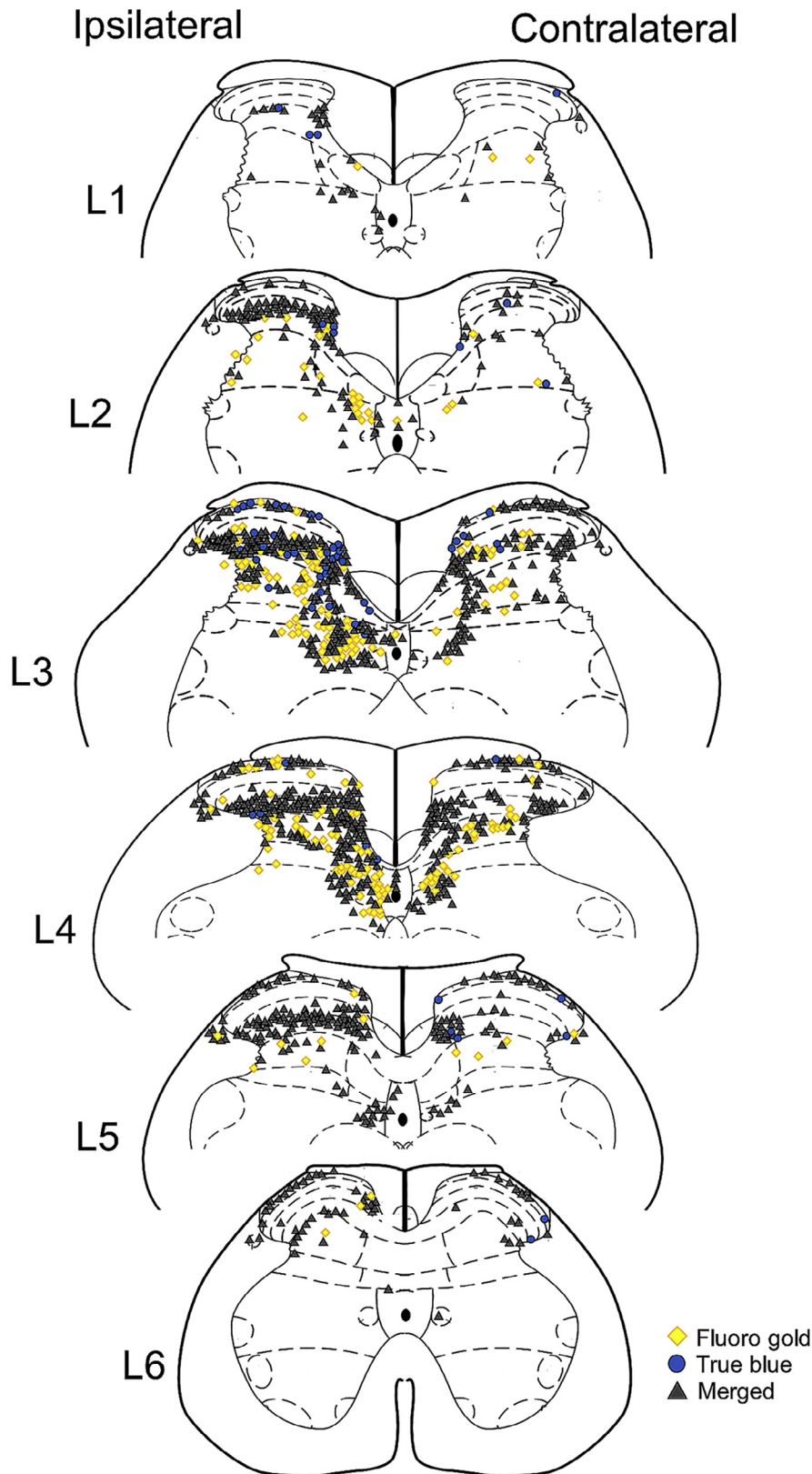


Fig. 3. Schematic illustration of the TB-, FG- and TB/FG-stained cells at spinal cord levels L1–L6. The plotted results were obtained from two representative rats. The blue circles represent the TB-positive cells, the yellow circles represent the FG-positive cells, and the black triangles represent the TB/FG-positive cells. Note the reduced number of labeled cells at L1 and L6. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

TB and FG antibody-labeled neurons in the spinal cord

After confirming that the spinal cord cells project to the GRA and the VPL using two different retrograde tracers, we decided to use the FG antibody to label the spinal cord sections that were injected with FG into the VPL for immunofluorescence staining.

Fig. 2C1, C2 and C3 shows examples of the TB-, FG- and TB/FG-positive spinal cord cells, respectively. The images of the cells that were stained with the FG antibody and an AlexaFluor 555-conjugated secondary antibody were observed through confocal microscopy. Briefly, the confocal microscopy results using the FG antibody also confirmed the presence of double-labeled and bilateral stained cells, as illustrated in Fig. 2D1, D2 and D3 in a mosaic composition of a lumbar spinal section. Fig. 2 also illustrates the location of the stained cells in the spinal cord.

Bilateral and bifurcated spinal cord cells

Using the confocal microscope (Zeiss Axiovert LSM 500 meta) with a multiphotonic laser (Chameleon ultra) and the appropriate filters, we captured mosaic images of the FG-, TB- and FG/TB-labeled cells in the spinal cord. These images are presented in Fig. 2E1, E2 and E3. It is important to note that TB- or FG-stained cells are present in both sides of the spinal cord; thus, we termed them bilateral and bifurcated cells because they send projections to the GRA and VPL. However, the number and distribution of these cells were not uniform along the lumbar segments, with the highest number of them in the L4 and L5 segments of the spinal cord. Six hundred sections from two rats were used to show the distribution and location of the bilateral and bifurcated spinal cord neurons along lumbar segments. This distribution is plotted in Fig. 3 using the drawings of the spinal cord provided by the 1998 Paxinos and Watson atlas. We observed the highest number of cells in layers 1 and 4, and in the lateral and medial portions of layers 5, 6 and 7.

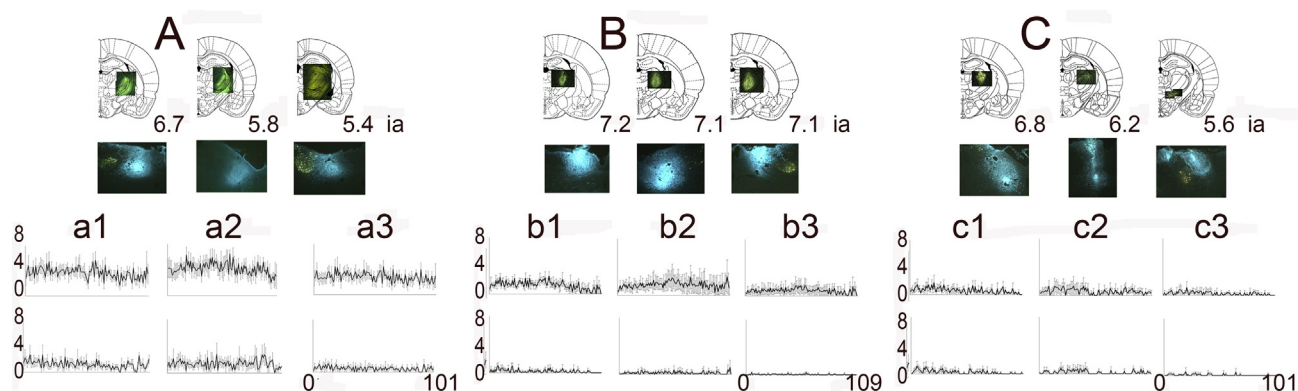


Fig. 4. Histological location of the thalamic FG injection and GRA-FG deposit. A: shows three anteroposterior schemas with large FG injections, giving the average amount (three rats) for a1 FG, a2 TB and a3 FG/TB-stained cells ipsilateral (upper graph) and contralateral (lower graph) to the GRA injection side. B: four schemas of thalamic medium injections and in the lower graphs the mean number of ipsilateral (upper graph) and contralateral (lower graph) b1 FG-, b2 TB- and b3 FG/TB-stained cells in the spinal cord. C: A histological example of smaller tracer deposits. Notice that the injection sites (the brightest) are surrounded by a local transport, giving the impression of larger injection sites. ia: distance from interaural plane.

Relationship between double-stained cells and tracer size and location of tracers

Three groups of three rats each were selected according to the size of injections in the thalamus and GRA regions.

The largest FG thalamic injection reached the rostral portion of the VPL from 6.70 to 5.40 anteroposterior plane, and the TB injection into the GRA included the ventral and the dorsal parts of the nucleus tractus solitarius (Sol). With these sizes of tracer deposits, FG- and TB-positive cells and double-stained FG/TB cells were found in superficial 1, 2; intermediate 4, 5; and deep 6, 7 spinal cord layers (Fig. 4A, a1, a2, a3). The superficial FG-positive cells seem related to the posterior thalamic nuclei (Po), while the intermediate cells seem related to the lateral dorsal ventral (LDV), medial (LDDM), anterior ventral (AV) and ventral lateral (VL) thalamic nucleus. The deeper FG cells were related to the VL, zona incerta (ZI) and reticular (RT) thalamic nucleus. The relationships between structures and stained spinal cord cells were corroborated with smaller tracer injections (Fig. 4B, C). The quantity of superficial TB-stained cells was related to the GRA and Sol participation and cells were less numerous in deeper laminae.

Smaller FG and TB injections let us to confirm the involvement of the above-described nuclei because less cells were stained in the spinal cord (Fig. 4B, b1, b2, b3 illustrates these results). The smaller deposits of tracers confirm the results but with less quantity of spinal cord stained cells (Fig. 4C, c1, c2, c3). Moreover, we observed that Sol involvement is related to the superficial spinal cord cells. In the three groups of rats, we observed stained cells with TB, FG and TB/FG in both sides of the spinal cord. Smaller and more anterior FG and TB deposits generated stained cells in both sides of spinal cord, thus we decided to confirm the presence of the bilateral and bifurcated spinal cord cells using another methodological approach.

Characteristics of cells recorded and antidromic or orthodromic cell responses to GRA or VPL electrical stimulation

We focused our experimental interest on wide dynamic range cells (WDR) due to their participation in nociception. In Fig. 5 an example of a WDR cell is plotted during electrical stimulation of its receptive field. The peri stimulus time histograms allowed us to specify the amount of A-beta-fiber (0–20 ms), A-delta-fiber (20–90 ms), and C-fiber (90–300 ms) action potentials produced by each train of 20 stimuli. The latencies and the estimated type of activated fiber were in agreement with previous reports (Urch and Dickenson, 2003).

Twenty-seven cells responded antidromically (Fig. 6) and 78 cells responded orthodromically (Fig. 7) to the electrical stimulation of the GRA. Seventeen cells responded antidromically and 43 responded orthodromically to the electrical stimulation of the VPL.

The antidromic spinal cord cells that were activated by the GRA stimulation had a mean latency of 9.85 ± 1.001 ms, which corresponded to a conduction speed of 10.96 ± 0.989 m/s. The antidromic cells that were activated by the VPL stimulation had a mean latency of 10.82 ± 1.255 , which corresponded to a conduction speed of 11.36 ± 1.239 m/s. Table 2 A shows the mean latency and estimated conduction speed for the antidromic cells responding only to GRA(-anti) or VPL(-anti) stimulation. Because different pathways, including cortical relays could mediate the orthodromic responses, the conduction speeds and latencies of these cells were not included in Table 2.

Antidromic branched cells respond to GRA and VPL electrical stimulation

Seven cells were antidromically activated by the electrical stimulation of both GRA and VPL (in Table 2 A GRA-anti-anti and VPL-anti-anti). The mean latency for GRA-activated cells and VPL-activated cells was 7.28 ± 1.39 ms and 13.71 ± 2.417 ms, respectively. Fig. 6 shows

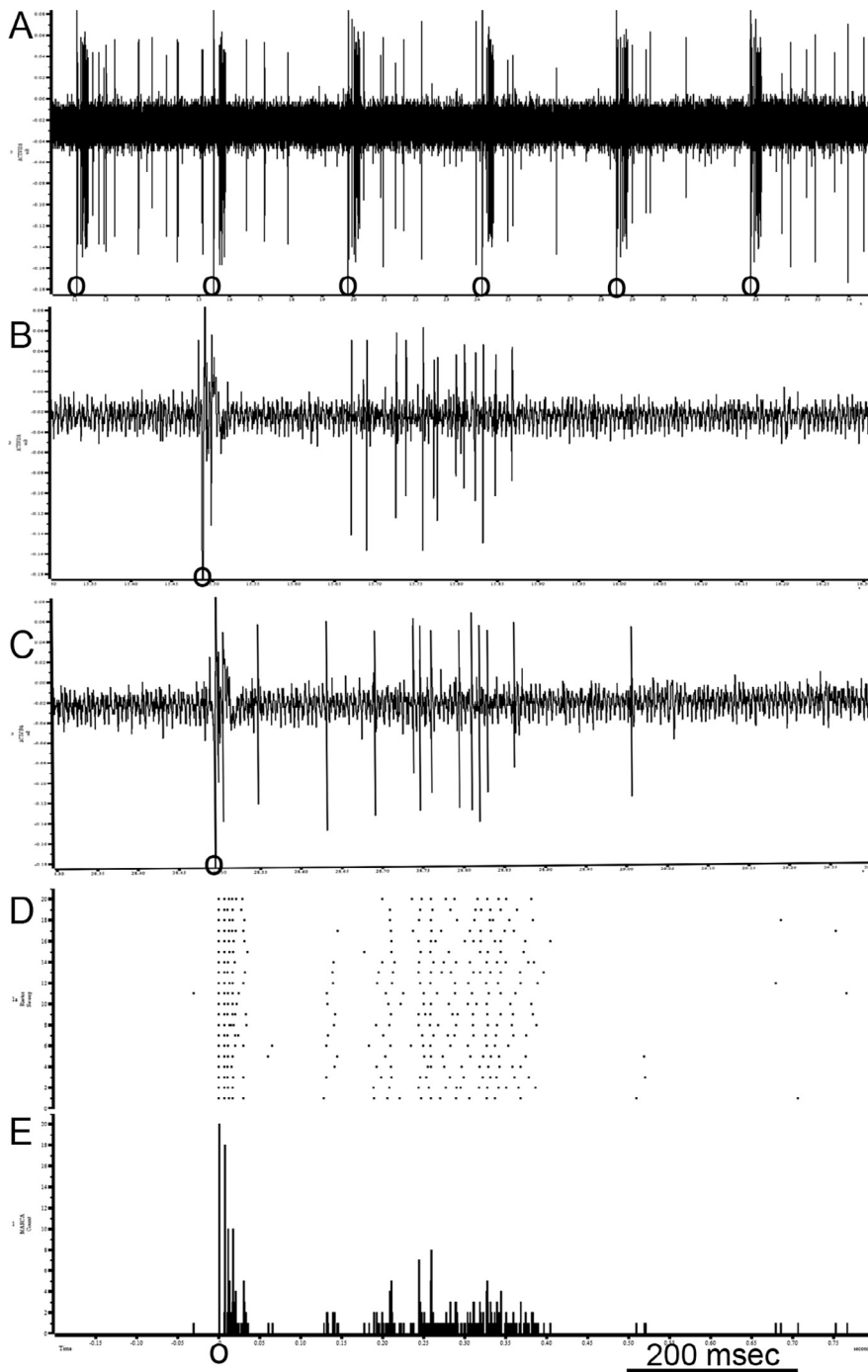


Fig. 5. WDR neuron responding to electrical stimulation of receptive field. In (A) six consecutive responses to electric stimulation are shown. In (B and C) single responses are plotted. In (D) raster display showing the consistence of the responses and in (E) the peristimulus time histogram of the digitalized cell action potentials. Notice that between 200 and 300 ms the C fibers were activated. Bin time 1 ms. 0 designates the stimulus artifact.

an example of an antidromic cell responding to electrical stimulation of both the GRA (Fig. 6A) and the VPL (Fig. 6B). It is important to notice that the cell activity illustrated in Fig. 6 is from a cell with the shortest latency and it was found in the superficial layers of the dorsal horn.

The mean conduction velocity was 13.689 ± 1.937 m/s for the cells projecting to the GRA and 9.629 ± 2.324 m/s for the cells projecting to the VPL. It is interesting to notice that the GRA and VPL latencies were significantly different ($t = 3.198$; $gl = 6$; $P < 0.01$), but the conduction velocities were not ($t = 2.27$; $gl = 6$; $P < 0.6$). These values are listed in Table 2B. Another characteristic of spinal cord cells that respond to the antidromic stimulation was that they can follow high frequencies of stimulation, which is described as a characteristic (Fuller and Schlag, 1976) of the antidromic cells responses as it is illustrated in Fig. 6C, D.

Location of the recorded cells. After the recording and collision test, neurons that exhibited antidromic responses to GRA or VPL electrical stimulation were marked by an iontophoretic pontamine blue injection. The mean depth of 17 localized recorded cells was $748.9 \pm 307.7 \mu\text{m}$; two examples of histological spinal cord sections showing the recording sites are provided in Fig. 8.

DISCUSSION

General remarks

Our results show a higher proportion of ipsilateral bifurcated projecting cells (4.6 cells) when compared to the contralateral side (1 cell), however further studies are needed to see if this result can be replicated while using the right side as a reference.

The application of retrograde tracers in the GRA and VPL and the use of three different approaches TB (GRA)/FG(VPL) and TB(GRA)/FR (VPL) with direct fluorescence or immunofluorescence procedures, let us to detect labeled neurons on both sides of the lumbar segments of the spinal cord. This result was

confirmed using the electrophysiological technique of antidromic activation, demonstrating that the ipsilateral GRA and the contralateral VPL electrical stimulations produced antidromic activities in the same spinal-cord cell. In brief, these results indicate a subset of L1-L6 spinal cord cells that project directly and bilaterally to the GRA and VPL.

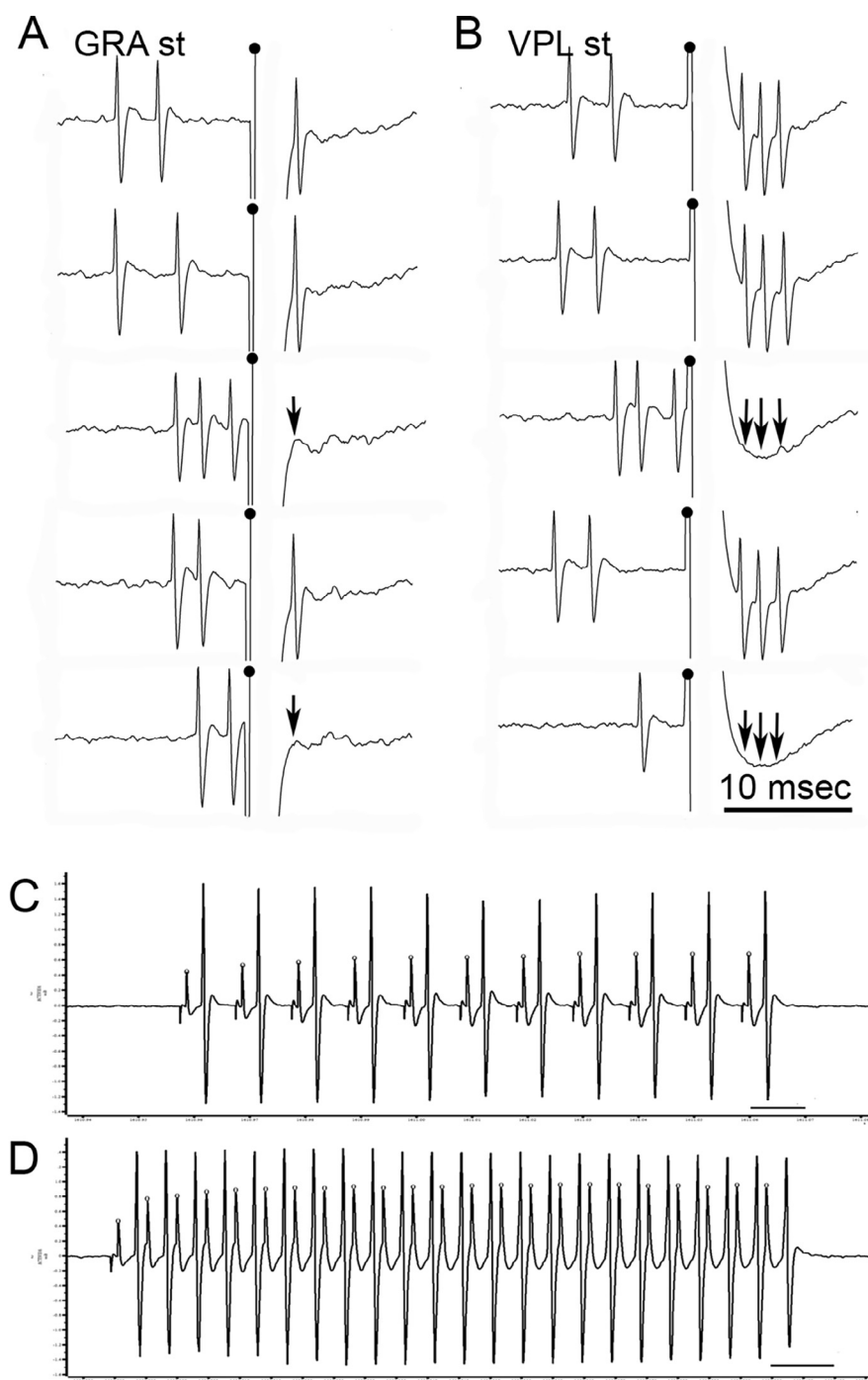


Fig. 6. Electrophysiological collision test. A single unit electrophysiological recording of a neuron responding to antidromic electrical stimulation of the GRA in (A) and VPL in (B) is illustrated. The arrows show the absence of an antidromic response due to the collision between the spontaneous and the antidromic action potential. In traces 1, 2 and 4, we can observe the fixed latency as a characteristic of the antidromic responses. Another characteristic of the antidromic-activated cells is their ability to follow high frequencies of stimulation. The recordings of spinal dorsal horn cells following 120 or 240 Hz stimulations are shown in (C and D) respectively.

Histological results

Extensive documentation has shown that the spinothalamic and dorsal column pathways are independent; however, the studies of [Apkarian and Hodge \(1989\)](#) and [Hodge and Apkarian \(1990\)](#) show a bilateral spinal cell

projection to the posterior ventral posterolateral thalamic complex. Using the retrograde tracer with HRP, these authors reported that at least 17% of the spinothalamic cells in the spinal cord were found ipsilateral to the thalamic injections. Also, [Kevetter and Willis \(1984\)](#) previously reported that the 19% of the lateral thalamus and the 38% of the medial thalamus received an ipsilateral spinal projection. These authors also reported that the 2% of spinothalamic cells project to both sides of the thalamus. Our findings confirm the ipsilateral and the contralateral spinal cord projection to the VPL and furthermore for the first time demonstrate that these spinal neurons have a bifurcated projection to the GRA. This bilateral and bifurcated projection involves the postsynaptic DC-ML and the spinothalamic pathway for the AL systems.

Importantly, when the VPL injection site was changed in the antero-posterior or in the medial-lateral plane, the number of double-labeled spinal cord projection cells changed possibly due to the topographical representation of the body in the VPL ([Mountcastle and Henneman, 1952](#)). However, in the current version only the lumbar segments of the spinal cord were analyzed.

In fact, the spinal cord projection neurons have been extensively studied, particularly the lamina I projections to the caudal ventrolateral medulla, lateral parabrachial area and periaqueductal gray matter. Most of the lamina I projection neurons have been identified when the TB deposits located in the deeper part of the GRA nuclei and reached the solitary tract, as has been suggested in previous descriptions ([Bester et al., 1997, 2000](#); [Gauriau and Bernard, 2002](#)). This lamina I projection has also shown to have bilateral projections ([Spike et al., 2003](#)). In another study, the ascending spinothalamic projections through the contralateral side, also have a component along the ipsilateral side ([Burstein et al., 1996](#)). In a

previous study ([Rojas-Piloni et al., 2008](#)), we used Diamidino Yellow (DY) and TB injections into the ipsilateral GRA and contralateral VPL, respectively, and we did not observe double-stained cells in the dorsal horn of

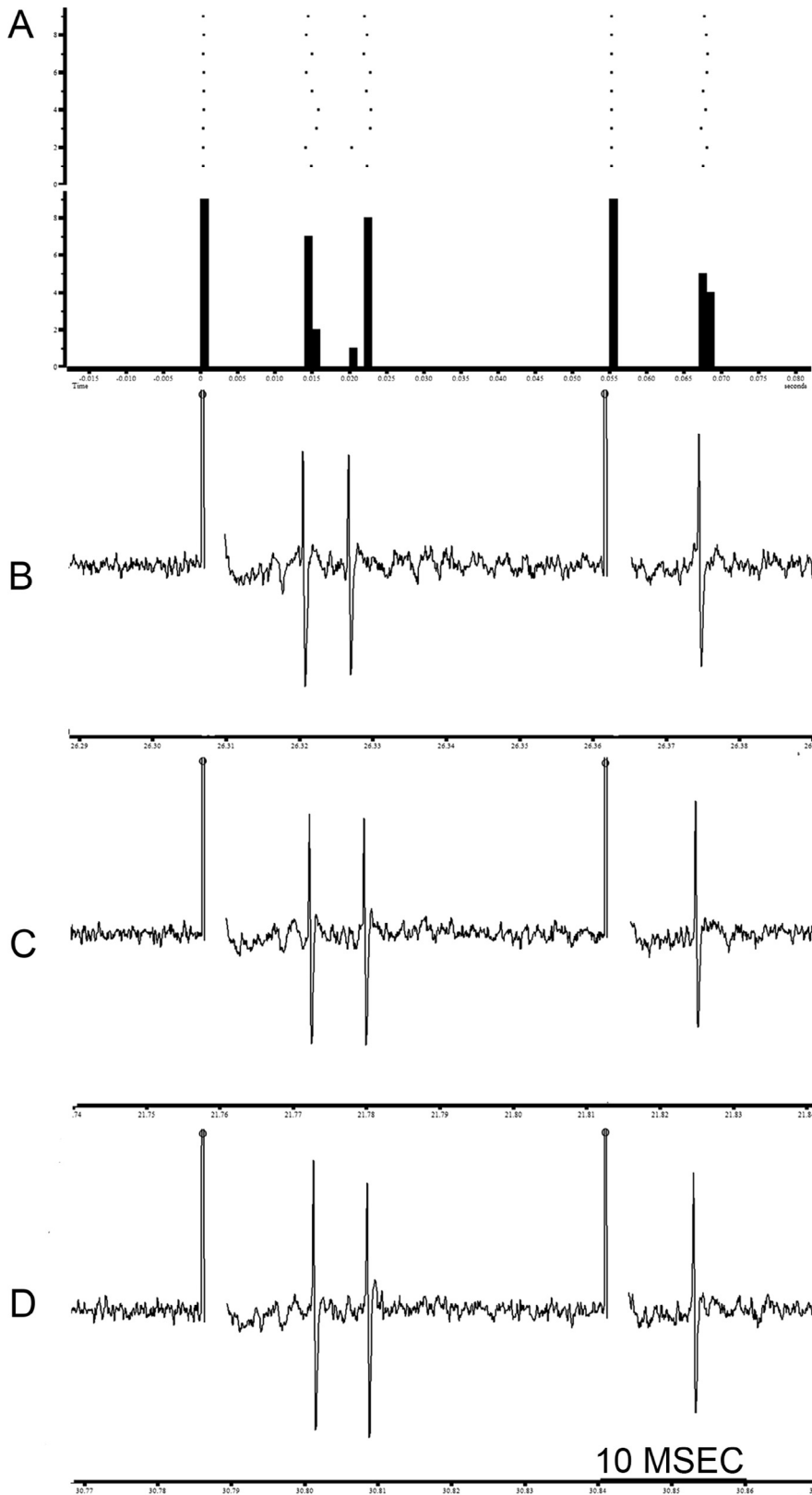


Fig. 7. Orthodromic responses to GRA and VPL electrical stimulation. In A, shows the raster display implemented with the digitalized signals for an orthodromic spinal cord cell responding to electrical stimulation of both GRA and VPL. In B, peri-stimulus time histogram of the stimulus artifact and the spike responses, the bin time was 1 ms. C, D and E individual recorded traces of the cell responses. The first responses on the left correspond to the GRA stimulation and the responses on the right correspond to the VPL stimulation. Notice the latencies variations in the responses, which allowed us to classify them as orthodromic.

the spinal cord. The main difference with our current results, is in the VPL injection sites, since the VPL tracer injection was more caudal (Ap 5.4 mm) and lateral (L 3.2 mm) compared to that in the present paper (Ap 6.1 and L 2.8 mm). This apparent discrepancy in the results due to the injection site locations only indicates the somatotopic and functional organization of the thalamus. Another interesting difference is that a particular cell population appears in the superficial layers of the spinal cord if the GRA injection involves the solitary tract, as described in the previous paragraphs. An important conclusion is that the neuroanatomical connections between the AL and DC-ML systems need to be conceptually restructured based on their functional interactions. Further studies need to address this subject.

Electrophysiological results

The recorded cells were characterized as wide dynamic range (WDR) cells due to their responses to different kinds of somatic stimulation intensities including noxious stimulation (see Fig. 5). Notice that between the antidromic activated cells by GRA and VPL we found one that was located in the superficial layers of the spinal cord. This cell displayed a characteristic double spike studied by Dickenson's works (Seagrove et al., 2004; Rojas-Piloni et al., 2007). This type of cells with double spontaneous and evoked spike activity is related with nociception and an enhanced response to noxious stimulation as wind-up (Dickenson, 1990). In fact, we focused our attention on WDR neurons because of their electrophysiological characteristics (e.g., they are projecting cells and are involved in nociception). It is interesting that Apkarian and Hodge (1989) proposed an ipsilateral spinal cord projection to the VPL that could be related to pain transmission in primates, but they did not provide any experimental support. Nevertheless, our present electrophysiological results support this hypothesis, namely that specific RF electrical stimulation activates the A delta and C fibers in the recorded spinal cord neurons.

Our present cell responses were orthodromic and antidromic to the stimulation of GRA or VPL, but the

Table 2. Latencies and conduction speeds values of antidromic activated cells. A, shows the mean and standard error values for the latencies (msec) and conduction speeds (m/s) for the antidromic (anti) responses produced by the GRA or VPL stimulation (as an example, GRA-anti-msec means that GRA stimulation produces an antidromic response and the value corresponds to the latency in msec. GRA-anti-m/s means that GRA stimulation produces an antidromic response and the value corresponds to the conduction speeds in m/s). The comparisons between the GRA and VPL latencies (msec) and conduction speeds (m/s) for the cells responding antidromically to GRA and VPL shown in B. The minimum value for significance was $P < 0.05$. Note that although the GRA vs. VPL anti-anti latency differences (msec) were significant, the differences in conduction speeds were not significant. These data are important for assuming that the axons that project to the VPL and the GRA have the same type of fiber. gl: liberty degrees; t: T value; NS: nonsignificant; **: $P < 0.05$; GRA: Gracilis nucleus; VPL: ventral posterolateral thalamic nucleus; anti: antidromic responses; anti-anti: antidromic responses for both GRA and VPL stimulation; msec: latency response in milliseconds; m/s: conduction speed

	N	Latency	m/s	
A				
GRA-anti	27	9.86 ± 1.00	10.96 ± 0.98	
VPL-anti	17	10.82 ± 1.25	11.36 ± 1.23	
GRA-anti-anti	7	7.28 ± 1.39	13.68 ± 1.93	
VPL-anti-anti	7	13.71 ± 2.41	9.62 ± 2.32	
		gl	t	P<
B				
GRAvsVPL-anti-anti-msec	6	3.198	0.019	**
GRAvsVPL-anti-anti-m/s	6	2.272	0.063	NS

most important finding was that some cells responded antidromically to both GRA and VPL electrical stimulation. This result confirms the existence of bifurcated spinal dorsal horn cells that project to the GRA and VPL. The antidromic activation technique has been widely used to demonstrate direct pathways, as well as cells with collaterals to different structures. The results shown in Table 2B demonstrate that the comparison between the conduction speeds of the antidromic spinal cord cells that responded to both GRA and VPL electrical stimulation were not significant, but the latencies were different, indicating that the spinal cord cells used the same size and type of bifurcated axons. Of course, the latencies were different because the distances were different.

Finally, the electrophysiological recordings using the collision test of spinal dorsal horn neurons during GRA and VPL electrical stimulation confirm our neuroanatomical results.

Functional considerations

Until now, it has been thought that the DC-ML system is related to the tactile and proprioceptive modalities, while the AL system is related to pain, temperature and crude touch. Moreover, the postsynaptic DC-ML system seems to be ipsilateral to the GRA or CUN nuclei at the medulla level, and the AL system is composed of several pathways, including the spino-thalamic, spino-mesencephalic and spinoreticular tracts. Strictly, our results only involved the spino-thalamic tract.

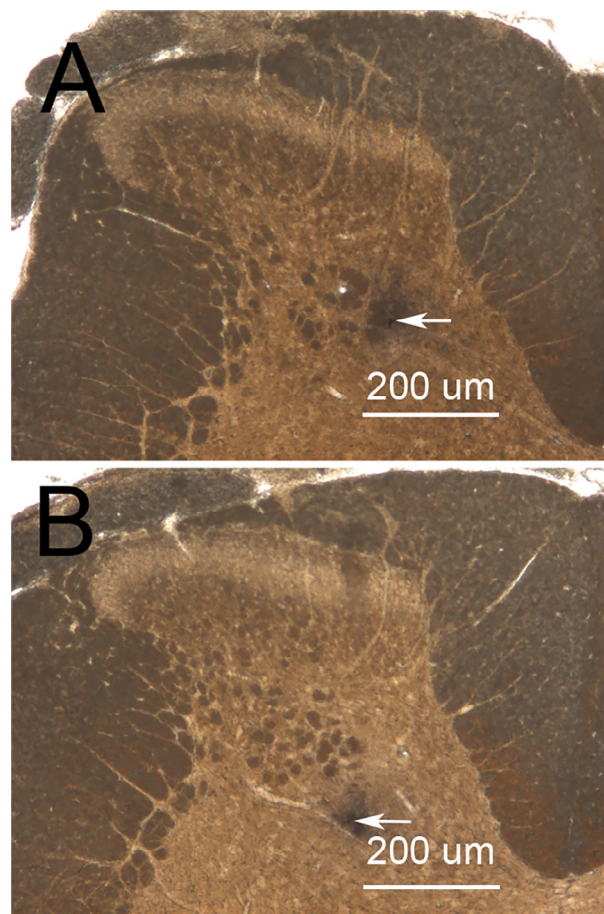


Fig. 8. Histological control of the recorded antidromic spinal cord cells. A section of the dorsal horn spinal cord at L4 is shown in (A) the white arrow points to the site of a recorded antidromic cell that responded to GRA and VPL electrical stimulation. The depth of the microdrive tip was 700 μm . Panel (B) shows another pontamine blue injection indicating the location of a recorded cell at a depth of 830 μm .

The present data show that single WDR spinal cord cells project bilaterally to the GRA and VPL, and it is possible that this neuronal arrangement serves to ensure that the alarm system of nociception generates a descending diffuse response. This last conclusion is supported by the finding that when the inhibitory paraventricular nucleus of the hypothalamus was stimulated, c-Fos activation was observed in the spinal cord, and the double and bifurcated cells were also c-Fos positive (Condés-Lara et al., 2015), providing negative feedback to reach a homeostatic state (Condés-Lara et al., 2009). This particular situation can explain the result showing that the threshold for nociceptive stimulation decreases on both sides of the body in neuropathic (Guilbaud et al., 1986; Desmeules et al., 1995) or carrageenan-induced (Guilbaud et al., 1986) pain models. In studies of human pain, the thalamic response has bilateral representation, likely reflecting generalized arousal in response to pain (Peyron et al., 2000). Additionally, a less common phenomenon reported in literature is the tactile hypersensitivity in the contralateral hind paw in experimental neuropathy. Some studies show contralateral tac-

tile allodynia from 10 to 120 days (Seltzer et al., 1990; Shir and Seltzer, 1990; Takaishi et al., 1996; Sinnott et al., 1999; Tabo et al., 1999). A 2-mm polyethylene cuff implanted around the sciatic nerve (Mosconi and Kruger, 1996) produces a decrease in the withdrawal threshold of both the ipsilateral and contralateral hind paws to normally innocuous tactile stimulation with von Frey hairs (Pitcher et al., 1999a,b). This evidence is of interest not only with respect to spinal cord pain representation, but also with respect to at least some forms of clinical chronic neuropathic pain (Kozin et al., 1976; Oaklander et al., 1998); it also suggests that altered sensory information processing may be established and maintained by bilateral peripheral nerve injury or inflammation (Bennett, 1999). Similarly, an increase in the contralateral neuronal firing rate was demonstrated in neuropathic rats (Pitcher and Henry, 2000).

Based on studies showing that tactile allodynia in experimental neuropathy is substantially decreased or abolished following spinal cord transection (Bian et al., 1998; Sung et al., 1998) or by an injection of lidocaine into the brain stem (Pertovaara et al., 1996). It has been argued that the greater tactile sensitivity of the nerve-injured hind paw is sustained via a supraspinal loop with tonic descending facilitating properties. In fact, the bilateral and bifurcated spinal cord projection neurons described in the present study could be responsible for this complex loop. But we don't know if this loop facilitates or inhibits the incoming afferent nociceptive and tactile information.

CONCLUSIONS

The results of the present study show that; (1) A sub-set of spinal dorsal horn cells is bifurcated and bilateral since they have projections to both main systems, the anterolateral and dorsal columns systems. (2) The unequal proportion of bifurcated cells regarding the body sides could be involved with the location of the stimulation source as a comparator. (3) The bilateral actions could interact with the diffuse endogenous inhibitory descending systems. (4) Further studies will be required to discern the real function of these novel interactions of the spinal cord projection cells.

CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to declare.

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